NAU DoMS POTW Problem #2: One in Front

Ian Schuster, Department of Physics, NAU
I affirm this solution is my own work and I did not use AI tools, the internet, or help from other people.

Problem Statement

> Find all integers $n \geq 1$ with the property that writing the digit 1 in front of the decimal representation of n^2 produces exactly n^3. For instance, if n^2 were 37, the new number would be 137.

My Solution

It's worth noting that, unless I'm really bad at using a calculator, $n^2 = 37\ s.t.\ n = \sqrt{37} \approx 6.083$ produces $n^3 \approx 225.1$, which, of course, is *not* 137, or really even close to it. With that out of the way, let's analyze the problem. What we know:

1. The domain of discourse is $\mathbb{D} = \{n | n \in \mathbb{N}\};; \mathbb{D} = \mathbb{N}$.
    a. n must be an integer greater than or equal to 1, and the set of natural numbers $\mathbb{N}$ (I am of the opinion that $0 \notin \mathbb{N}$) is a set that satisfies those conditions.
2. n^2 and n^3 are themselves $\in \mathbb{N}$, because $\mathbb{N}$ is closed under multiplication.
3. n^2 and n^3, in decimal representation, are related by the stated relationship.

So, the question then becomes how we can re-phrase this stated relationship into something we can play around with. I think the first thing I noticed is that n^3 must have *exactly* one more digit than n^2 to even be in consideration, as otherwise, "adding" a single digit (1) to the front of n^2 would not even produce the same number of digits as in n^3. How can we express this?

- $floor(log_{10}(x)) \stackrel{\text{def}}{=}$ The number of powers of 10 that fit into x. That is, if x is any two digit number, 1, if x is a three digit number, $floor(log_{10}(x)) = 2$, etc. Notationally, $floor(a) = \lfloor x \rfloor$. Note that the "floor" function here is a function that takes an element of $\mathbb{R}$ and returns the corresponding integer "part" of that real number. So, $\lfloor 2.56589 \rfloor = 2$. $digits(a) \stackrel{\text{def}}{=} \lfloor a \rfloor + 1$. So, if $\lfloor a \rfloor = 1$, $digits(a) = 2$ (this, then, means that $digits(log_{10}(x))$ is a function which returns the number of digits in a number x).

With this, we can say that $digits(log_{10}(n^3)) - digits(log_{10}(n^2))$ *must* equal 1 — if this were not the case, the condition "n^3 must have exactly one more digit than n^2" would not be true. If we could simplify the difference of $digits()$ into a $digits()$ of a difference, we could then take advantage of some log properties.

What does $digits(x) - digits(y)$ resolve to?
$digits(x) = \lfloor x \rfloor + 1, digits(y) = \lfloor y \rfloor + 1$. So, we're really asking:
What does $\lfloor x \rfloor + 1 - (\lfloor y \rfloor + 1) = \lfloor x \rfloor - \lfloor y \rfloor$ resolve to?
Note that we want to get it into some form that has a $\lfloor x - y \rfloor$, so we can take advantage of log properties

Consider two cases: $x, y \in \mathbb{N}_0$, $x \vee y \in \mathbb{N}_0$ (both still within $\mathbb{R} > 0$). Note that I am imposing the condition that $\lfloor\rfloor$ is a subsetting operation which takes an element of $\mathbb{R} \geq 0$ and returns an element of $\mathbb{N}_0$.

1. If x is a natural number, and y is a natural number, $\lfloor x \rfloor = x$, $\lfloor y \rfloor = y$, so $\lfloor x \rfloor - \lfloor y \rfloor = x - y$. Let $z \stackrel{\text{def}}{=} x - y$. Iff $x \geq y$, $z \in \mathbb{N}_0$, and then, $\lfloor z \rfloor$, of course, is equal to z, so $x - y = \lfloor z \rfloor$, $\lfloor x \rfloor - \lfloor y \rfloor = \lfloor z \rfloor$, $\lfloor x \rfloor - \lfloor y \rfloor = \lfloor x - y \rfloor$.

If $x, y \in \mathbb{N}_0$, and iff $x > y$, $\lfloor x \rfloor - \lfloor y \rfloor = \lfloor x - y \rfloor$.

2. If one, or both, of x, y are *not* natural numbers, we can describe x, y in the following terms: $x = n + q$, $y = m + r$, where $n, m \in \mathbb{N}_0$, $q, r \in \mathbb{R}\ s.t.\ 0 \leq q, r < 1$. Note that $q, r \geq 1$ isn't strictly disallowed, but we can just absorb any integer parts of q, r into n or m, respectively (that is, e.g. if $n = 1, q = 1.3$, just let $n \to n' = 2, q \to q' = 0.3$). $\lfloor x \rfloor$ is logically just n (this is a direct result of how the floor function works), and $\lfloor y \rfloor$ is logically just m, and $\lfloor x \rfloor - \lfloor y \rfloor = n - m$. $\lfloor x - y \rfloor$, on the other hand, is equal to $\lfloor n + q - (m + r) \rfloor = \lfloor n - m + q - r \rfloor$. Let $o \stackrel{\text{def}}{=} n - m$, and, iff $n \geq m$, $o \in \mathbb{N}_0$. Let $p \stackrel{\text{def}}{=} q - r$. Let's look at two sub-cases (a). $q \geq r$, and (b). $q < r$:
    a. If $q \geq r$, we know that $0 \leq p < 1$. Note that, even though p is positive in this case, it must always be less than 1 and greater than 0, because q, r are always less than 1 and not negative. In this case, $\lfloor o + p \rfloor = o$ (the same way that $\lfloor n + q \rfloor = n$). Because of how o and p are defined, $o + p = n - m + q - r = (n + q) - (m + r) = x - y$. So, $\lfloor o + p \rfloor = \lfloor x - y \rfloor$. o, of course, is just $n - m$, which we know to be equivalent to $\lfloor x \rfloor - \lfloor y \rfloor$. So, in this case, $\lfloor x \rfloor - \lfloor y \rfloor = \lfloor x - y \rfloor$, again.
    b. If $q < r$, we know that $-1 < p < 0$. We can perform a similar transform to that described with n and m. Let $o \to o' = o - 1$, and $p \to p' = p + 1$. Note here that, for o' to satisfy the sufficient condition of being an element of $\mathbb{N}_0$, $o \geq 1$. Then, $\lfloor o + p \rfloor = \lfloor o' + p' \rfloor$. $|p|$ must always be less than 1 and greater than 0 in this case, because q, r are always less than 1 and not negative (actually, this is true of case a. as well, but because the absolute value of a positive number is just that number, we can say $|p| = p$ in that case). If $0 < |p| < 1$, we know that $0 < p' < 1$. Then, $\lfloor o' + p' \rfloor = o'$ (this is just a result of the floor function),

and $\lfloor o + p \rfloor = o' = o - 1$. We already know $\lfloor o + p \rfloor = \lfloor x - y \rfloor$, and $o = \lfloor x \rfloor - \lfloor y \rfloor$. So, in this case, $\lfloor x \rfloor - \lfloor y \rfloor = \lfloor x - y \rfloor + 1$.

Note that, if $\lfloor x \rfloor - \lfloor y \rfloor = \lfloor x - y \rfloor$, x, y are either elements of the natural numbers with $x > y$, or the difference between the non-integer parts of x, y, named q, r (respectively) is greater than or equal to 0. If $\lfloor x \rfloor - \lfloor y \rfloor = \lfloor x - y \rfloor + 1$, the difference between q, r is less than 0.

So, $digits(log_{10}(n^3)) - digits(log_{10}(n^2) = \lfloor log_{10}(n^3) - log_{10}(n^2) \rfloor = \lfloor log_{10}(\frac{n^3}{n^2}) \rfloor$
$= \lfloor log_{10}(n) \rfloor$,

or

$digits(log_{10}(n^3)) - digits(log_{10}(n^2)) = \lfloor log_{10}(n^3) - log_{10}(n^2) \rfloor + 1 = \lfloor log_{10}(n) \rfloor + 1$

and which one we use is dependent on whether the q, r of $log_{10}(n^3)$ and $log_{10}(n^2)$ (respectively) is greater than/equal to or less than 0. Note that $log_{10}(n^3, n^2) \in \mathbb{N}_0$ iff $n = 10^k \; s.t. \; k \in \mathbb{N}$, so case one (above) doesn't apply very often. Let's analyze both of the cases:

(a). Because $digits()$ must equal 1, we know that $1 \leq log_{10}(n) < 2$ (otherwise, $\lfloor log_{10}(n) \rfloor$ would not return 1). The condition $1 \leq log_{10}(n)$ means, through log-exponent transform (that is, $log_a(b) = c \leftrightarrow a^c = b$), $10 \leq n$, and the condition $log_{10}(n) < 2$ means (also through log-exponent transform) $n < 100$. So, for n to even be under consideration for condition 3, under case a, $10 \leq n < 100$.

(b). Because $digits()$ must equal 1, we know that $0 \leq log_{10}(n) < 1$ (otherwise, $\lfloor log_{10}(n) \rfloor + 1$ would not return 1). The condition $0 \leq log_{10}(n)$ means, through log-exponent transform, $1 \leq n$, and the condition $log_{10}(n) < 1$ means (also through log-exponent transform) $n < 10$. So, for n to even be under consideration for condition 3, under case b, $1 \leq n < 10$.

We can determine which of these two we need to consider if we can determine q, r, so let's take a stab at that. $log_{10}(n^3) = 3log_{10}(n)$, and $log_{10}(n^2) = 2log_{10}(n)$. Taking our difference, we get $3log_{10}(n) - 2log_{10}(n) = log_{10}(n)$ (which is what we'd expect). Let $3log_{10}(n) = n + q$
$2log_{10}(n) = m + r$, $log_{10}(n) = l + s$. Then, $(n + q) - (m + r) = l + s$;
$o + p = l + s. \; p = q - r$, then, is just the non-integer part of $log_{10}(n)$, and $o = \lfloor log_{10}(n) \rfloor$.

As a quick non-sequitur, consider that, in case b, I said that $o \geq 1$, else $o' \notin \mathbb{N}_0$. This is not entirely true; if we let $o < 1$, $o' < 0$. However, we can absorb this negative into p by the

following transformation. Recognize that $\lfloor o' + p' \rfloor = o'$, multiply both sides by a negative one, and we get $\lfloor -o' - p' \rfloor = -o'$. $-o'$ is, of course, just some $\in \mathbb{N}_0$, so let $o' \to o'' = -o'$ (and o'' becomes the element of $\mathbb{N}_0$), and p (or, in this case, p') is allowed to be negative, so let $p' \to p'' = -p'$. We then have $\lfloor o'' + p'' \rfloor = o''$, and see that our relationship can still hold if $o < 1$. Note that $o < 1$ means $o = 0$ (o is still bound by $\mathbb{N}_0$, even if o' is not).

This is where I'm going to invoke perhaps a non-following idea. Realize that, if our problem statement relationship is to hold true, $log_{10}(n^3 - n^2) \in \mathbb{N}$ (adding a one to the front of our number is the same as adding a power of 10 to that number; that is $n^3 - n^2 = 10^k$). Log-exponent transform says $log_{10}(n^3 - n^2) = w \to 10^w = n^3 - n^2$. Let's evaluate this function for values of w until we get some n greater than our bounds on n. Note that I elected to not include the simplification of the cubic formula here to prevent an extreme amount of clutter, but I did calculate it by hand, and am happy to provide said simplification if needed.
For

w =	10^w =	n =
1	10	~2.54
2	100	5
3	1000	~10.3
4	10000	~21.9
5	100000	~46.8
6	1000000	~100.3

Because $w = 6$ provides a value greater than our bounds, we know that these are all the possible n which satisfy. Of course, only one of these n is an integer, satisfying the conditions of the greater problem. As such, we know that the only integer which satisfies our problem statement is $n = 5$.

As a quick check, if $n = 5, n^2 = 25, n^3 = 125$.